

ARIAN RAHMAN ADITTA

Electronics & Communication Engineering Graduate | RFIC • VLSI/EDA • RF/Microwave

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[LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

RESEARCH INTERESTS

- RF/mmWave Integrated Circuits • Analog/RF IC Design • RFIC Design Automation • ML-Assisted Circuit Design
- VLSI/EDA

EDUCATION

B.Sc. Engineering in Electronics and Communication Engineering

[2022 - 2026]

Khulna University of Engineering and Technology, Khulna, Bangladesh

CGPA: 3.71/4.00

Final four term GPAs: 3.89, 4.00, 4.00, 3.88

Selected Coursework: VLSI Design & Nanotechnology, Analog Electronics, Microwave Engineering, Digital Communications, Antenna Engineering, Signals & Systems, Digital Signal Processing, Microprocessors & Microcomputers, Digital Electronics & Logic Circuits, Machine Learning

RESEARCH EXPERIENCE

Open-Source RFIC Design and Verification of a 28-GHz Two-Stage Class-A SiGe Power Amplifier

Undergraduate Thesis, Department of ECE, KUET | Supervisor: Aysha Tabassum | 2026

- Designed and simulated a two-stage 28-GHz Class-A SiGe power amplifier using the IHP SG13G2 130-nm SiGe BiCMOS Open PDK and an open-source RFIC workflow based on Qucs-S, ngspice/Xyce, KLayout, and EM-aware modeling.
- Developed a progressive ideal-passive → PDK-aware → EM-aware realization workflow; the final process-aware configuration achieved 17.4 dB gain at 28 GHz, $S_{11}/S_{22} \approx -13$ dB, and $K = 12$, while quantifying the impact of physical passive realization.
- Explored machine-learning-assisted design-space screening and optimization; currently extending the work toward layout/GDSII-level realization, with a thesis-derived manuscript submitted to ICCIT 2026.

Leakage-Aware RTL-to-QoR Prediction for Early Area and Timing Estimation

Research Collaboration with Zakia Sultana, Ph.D. Candidate, MSCAD Laboratory, University of Arkansas | Ongoing

- Developing machine-learning methods to predict mapped cell area and pre-layout timing directly from RTL before full implementation.
- Working on source provenance, duplicate/near-duplicate control, leakage-aware grouped evaluation, RTL structural feature engineering, and reproducible model-development pipelines.
- Investigating size-aware prediction, model explainability, coverage/extrapolation analysis, and out-of-distribution awareness for more reliable early-stage QoR estimation.

GNN-SABLE: GNN-Guided VLSI Floorplanning with Constraint-Aware Refinement

Independent Research | 2026–Present

- Developing a learning-guided VLSI floorplanning framework that combines GNN-based placement guidance with B*-tree simulated annealing, legalization and exact-evaluated refinement.
- Investigating trade-offs among wirelength, placement compactness, and physical-design legality under boundary, grouping and related structural constraints.
- Building a reproducible experimental workflow with controlled baselines, ablation studies, leakage-aware data handling, legality checks and stage-level evaluation for publication-quality and reproducible benchmarking.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Publications

- [1] **A. R. Aditta**, P. K. Das, U. R. Arshee, and A. B. M. Aowlad Hossain, "SenseStick: A Bangla-English Edge-AI Cane for Road-Safety and Biomedical Reliability Monitoring," *IEEE BECITHCON*, 2026.
- [2] P. K. Das, S. M. R. Islam, and **A. R. Aditta**, "LMatchNet-V: Physics-Seeded Learning for Vendor-Aware RF L-Section Matching with Adaptive Safe-Library Search," *IEEE MARSCON*, 2026.

[3] M. A. Jahin, S. Soudeep, **A. R. Aditta**, M. F. Mridha, N. Fahad, and M. J. Hossen, "Vision Transformers for End-to-End Quark-Gluon Jet Classification from Calorimeter Images," *Communications in Computer and Information Science*, vol. 2640, Springer, pp. 135–150, 2025. DOI: 10.1007/978-981-95-0988-1_10

Manuscript Submitted

[4] **A. R. Aditta** and A. Tabassum, "Progressive PDK-to-EM Realization of a 28 GHz Two-Stage Class-A SiGe Power Amplifier in an Open-Source RFIC Flow," submitted to *ICCIT 2026*.

Manuscript Under Revision

[5] **A. R. Aditta**, U. R. Arshee, S. M. R. Islam, and N. Taaha, "Transformer-Based Modeling of DNA Enhancers: A Controlled Evaluation of DNABERT and Longformer for Identification and Strength Prediction," manuscript under revision for journal resubmission, 2026.

TECHNICAL SKILLS

- **RFIC / Circuit / EM:** Cadence Virtuoso, Qucs-S, ngspice, Xyce, KLayout, openEMS, CST Studio Suite, LTspice, IHP SG13G2 Open PDK
- **VLSI / EDA / RTL:** Verilog, Yosys, OpenSTA, RTL analysis, synthesis/STA workflows, B*-tree floorplanning, physical-design optimization
- **Programming & Scientific Computing:** Python, C, C++, MATLAB, NumPy, Pandas
- **Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, GNNs, CNNs, tree-based models, regression/classification, model evaluation
- **Embedded Systems:** Raspberry Pi, ESP32, Arduino, OpenCV
- **Research Tools:** Git, LaTeX, Zotero

HONORS & AWARDS

Dean's Award, 3rd & 4th Year, Khulna University of Engineering & Technology

- Awarded for outstanding academic performance in consecutive senior academic years.

Technical Scholarship, Khulna University of Engineering and Technology (KUET)

- Awarded for academic performance during the first three undergraduate years.

ACADEMIC & RESEARCH LEADERSHIP

President, KUET Research Society, ECE

[2025-2026]

- Organized a department-wide pre-mid thesis competition involving 3rd- and 4th-year ECE students to promote research presentation, peer learning and early thesis engagement.
- Coordinated online graduate-study and research guidance sessions with KUET alumni pursuing M.S./Ph.D. degrees abroad, helping current students understand research preparation, graduate applications, and academic career pathways.

General Secretary, KUET Debating Society

[2023-2026]

- Led organizational activities and competitive debating initiatives, developing extensive experience in communication, team leadership, public speaking, team coordination and event management.

RELEVANT PROFESSIONAL EXPERIENCE

Campus Ambassador, Ulkasemi

[Aug 2024 – Sep 2025]

- Supported KUET–Ulkasemi semiconductor-industry engagement and student outreach; helped organize a five-day career and management training program for final-year students.

Data Analyst Intern, WEGRO

[Mar 2025 – Jul 2025]

- Analyzed business data, created decision-oriented visualizations, and contributed to data-driven process improvement in an agri-fintech startup.

ACADEMIC REFERENCES

Dr. A. B. M. Aowlad Hossain

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Dr. Sheikh Md. Rabiul Islam

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